1. Vector Space.

(a) A vector space V over $\mathbb{F}$ is finite-dimensional if $\exists k \geq 0$, such that $\exists$ a linear transformation $T: \mathbb{F}^k \to V$ is surjective.

(b) Proof by Contradiction.

Assume V is finite-dimensional. then according to definition, $\exists k \geq 0$ s.t. $T: \mathbb{F}^k \to \mathbb{F}[x]$ is surjective.

Take $\forall$ element x in $\mathbb{F}^k$. $x = (a_1, \cdots, a_k)$. we have,

$$T(a_1, \cdots, a_k) = T(a_1 e_1 + \cdots + a_k e_k) = T(a_1 e_1) + \cdots + T(a_k e_k) = \sum_{i=1}^{k} a_i T(e_i)$$

where $e_i = (\underbrace{0, \cdots, 0}_{i-1}, 1, \underbrace{0, \cdots, 0}_{k-i})$.

Then $\mathrm{Im}(T) = \mathrm{span}\{T(e_1), \cdots, T(e_k)\}$ i.e. $\forall v \in V$. is a linear combination of $T(e_1), \cdots, T(e_k)$. Since each $T(e_i)$ is a polynomial. Let $N = \max_{1 \leq i \leq k} \{\deg(T(e_i))\}$ is a finite number. However. $x^{N+1} \in \mathbb{F}[x]$ and $x^{N+1} \notin \mathrm{Im}(T)$.

$\Rightarrow T$ is not onto. Contradict!

Therefore, $\mathbb{F}[x]$ is not finite-dimensional.

(c) X is a finite set. Denote $|X| = k$. $X = \{x_1, \cdots, x_k\}$.

Then Consider $T: \mathbb{F}^k \to \mathbb{F}(X)$. we want to find a T s.t. T is onto.

Take $\forall f \in \mathbb{F}(X)$ Let $x = (f(x_1), \cdots, f(x_k))$ and $T(x) = T(f(x_1), \cdots, f(x_k)) = f$.

Then such onto map exists! $\Rightarrow V$ is finite-dimensional.

2.

(a) A sequence $0 \to U \xrightarrow{\iota} V \xrightarrow{\varsigma} W \to 0$ is short exact sequence if $\forall$ Linear Transformation there, the kernel equal the image of previous map. ι is injective, ς is surjective.

(b) $0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{\varsigma} W \longrightarrow 0$

$\iota: \iota(u) = u$ where $u \in U$.

$\forall v \in V$, $\exists u \in U, w \in W$ s.t. $v = u + w$. $\Rightarrow \varsigma(v) = \varsigma(u+w) = w$

$\Rightarrow \varsigma: \varsigma(u+w) = w$. where $u \in U, w \in W$

(c) $0 \longrightarrow |F(X) \xrightarrow{\iota} |F(Y) \xrightarrow{\varsigma} |F(Y/X) \longrightarrow 0$.

$\iota: |F(X) \longrightarrow |F(Y)$

$u \longmapsto u$

$\varsigma: |F(Y) \longrightarrow |F(Y/X)$

$$\varsigma(u) = \begin{cases} 0, & u \in |F(X) \\ u, & u \notin |F(X) \end{cases}$$

Obviously, ι is injective. For $\forall |F(u)$ where $u \in Y/X$. we have $u \in Y \xRightarrow{\varsigma} \varsigma(|F(u)) = |F(u)$

$\Rightarrow \varsigma$ is surjective.

Take $\forall |F(u)$ where $u \in X$. we have $\varsigma(|F(u)) = 0 \Rightarrow \ker(\varsigma) = \mathrm{Im}(\iota)$

3. Dimension.

(a)

$\exists$ a unique function $\dim : \mathrm{Vect}_{\mathbb{F}}^{fd} \to \mathbb{N}$ such that $\forall$ short exact sequence

$0 \to U \xrightarrow{i} V \xrightarrow{q} W \to 0$ has the property $\dim(V) = \dim(U) + \dim(W)$

(b)

Lemma: $\dim(\mathbb{F}^n) = n$.

Proof by induction:

base case: $k = 1$ $\dim(\mathbb{F}) = 1$.

Inductive Hypothesis: Assume $\dim(\mathbb{F}^k) = k$, want to prove $\dim(\mathbb{F}^{k+1}) = k+1$.

$0 \to \mathbb{F} \xrightarrow{i} \mathbb{F}^{k+1} \xrightarrow{q} \mathbb{F}^k \to 0$

$i : \mathbb{F} \to \mathbb{F}^{k+1}$

$x \mapsto (x, 0, \cdots, 0)$

$q : \mathbb{F}^{k+1} \to \mathbb{F}^k$

$(x_1, \cdots, x_{k+1}) \mapsto (x_2, \cdots, x_{k+1})$

Then this is a s.e.s. Using the proposition above, $\dim(\mathbb{F}^{k+1}) = \dim(\mathbb{F}) + \dim(\mathbb{F}^k) = k+1$.

$\Rightarrow \dim(\mathbb{F}^n) = n$.

Because $M_n(\mathbb{F})$ is isomorphic to $\mathbb{F}^{n^2}$. $\Rightarrow \dim(M_n(\mathbb{F})) = \dim(\mathbb{F}^{n^2}) = n^2$.

(c)1. $0 \to U \xrightarrow{i} V \xrightarrow{q} W \to 0$

$i : A_n(\mathbb{F}) \to M_n(\mathbb{F})$

$m \mapsto m$.

$q : M_n(\mathbb{F}) \to S_n(\mathbb{F})$

$q(m) = \begin{cases} 0, & m \in A_n(\mathbb{F}) \\ m, & m \notin A_n(\mathbb{F}) \end{cases}$

Since $M_n(\mathbb{F}) = \frac{M_n(\mathbb{F}) + M_n(\mathbb{F})^T}{2} + \frac{M_n(\mathbb{F}) - M_n(\mathbb{F})^T}{2}$

$\left(\frac{M_n(\mathbb{F}) + M_n(\mathbb{F})^T}{2}\right)^T = \frac{M_n(\mathbb{F}) + M_n(\mathbb{F})^T}{2} \in S_n(\mathbb{F})$ $\left(\frac{M_n(\mathbb{F}) - M_n(\mathbb{F})^T}{2}\right)^T = -\frac{M_n(\mathbb{F}) - M_n(\mathbb{F})^T}{2} \in A_n(\mathbb{F})$

$\Rightarrow M_n(\mathbb{F}) = S_n(\mathbb{F}) \oplus A_n(\mathbb{F})$. According to 2(b), such constructed s.e.s holds.

2. Because the diagonal of $A_n(\mathbb{F})$ should be all 0 but diagonal of $S_n(\mathbb{F})$ can be anything.

$\Rightarrow \dim(S_n(\mathbb{F})) = n + \dim(A_n(\mathbb{F}))$. Since $\dim(M_n) = \dim(S_n) + \dim(A_n) = n^2$.

$\Rightarrow \dim(A_n(\mathbb{F})) = \frac{n(n-1)}{2}$ $\dim(S_n(\mathbb{F})) = \frac{n(n+1)}{2}$